

Homework Assignment 2

Problem 1:

1Mbps link.

Data rate = 100Kbps

$p = 0.1$

1Mbps replaced by 1Gbps

$$(a) N = \frac{\text{total transmission rate}}{\text{Rate of data generation}}$$

$$= \frac{1 \text{ Gbps}}{100 \text{ kbps}}$$

$$= \frac{1000 \times 1000 \times 1000 \text{ bps}}{100 \times 1000 \text{ bps}}$$

$$= 10,000 \text{ users.}$$

(b) Probability that more than N users are sending data

$$= M \sum_{n=N+1}^M n = N+1 \left(M p^n (1-p)^{M-n} \right)$$

Problem 2:

Packet Length = L

d_i = length

s_i = propagation speed.

R_i = transmission rate.

for $i = 1, 2$.

Packet size = 1000 bytes

prop speed = 2.5×10^8 m/s.

transmission rate = 1 Mbps.

delay = 1 msec.

length = 4000 km.

(fits)

last link = 1000 km

1) Transmission Delay on Link 1:

$$\frac{L}{R_1} = \frac{8000 \text{ bits}}{1 \times 10^6 \text{ bits/s}} = 0.008 \text{ seconds} = 8 \text{ ms.}$$

Propagation Delay on Link 1:

$$\frac{d_1}{s_1} = \frac{4 \times 10^6 \text{ m}}{2.5 \times 10^8 \text{ m/s}} = 0.016 \text{ s} = 16 \text{ ms}$$

$$\begin{aligned}\text{Transmission Delay in last link} &= \frac{8000 \text{ bits}}{1 \times 10^6 \text{ bits/s}} \\ &= 0.008 \text{ s} \\ &= 8 \text{ ms}\end{aligned}$$

Propagation delay on ^{last} Link 2 :

$$\frac{d_2}{s_2} = \frac{1 \times 10^4 \text{ m}}{2.5 \times 10^8 \text{ m/s}} = 0.004 \text{ s} = 4 \text{ ms}$$

$$\begin{aligned}\text{Total end to end delay} &= (8 \text{ ms} + 16 \text{ ms}) + 1 \text{ ms} + (8 + 4) \text{ ms} \\ &= 37 \text{ ms}\end{aligned}$$

Problem 3:

$$\text{Message} = 7.5 \times 10^6 \text{ bits}$$

$$\text{Each link} = 1.5 \text{ Mbps}$$

(a) Without message segmentation.

Total time = ?

$$\begin{aligned} \text{Time from source to first packet switch} &= \frac{7.5 \times 10^6}{1.5 \times 10^6} \\ &= 5 \text{ seconds} \end{aligned}$$

$$\text{Total time} = 5 \text{ seconds} \times 3 \text{ hops} = 15 \text{ seconds}$$

(b) Message segmented into 5000 packets

with each packet = 1500 bits

Second packet?

$$\begin{aligned} \text{Time to send first packet} &= \frac{1500}{1.5 \times 10^6} \\ &= 0.001 \text{ seconds.} \end{aligned}$$

$$\begin{aligned} \text{Time when second packet is received} &= 2 \times 0.001 \\ &= 0.002 \text{ seconds.} \end{aligned}$$

(c) Total time when message segmented into 5000 packets
 $= 1\text{ms} \times 3 \text{ hops} = \cancel{3\text{ms} \times 5} + (4999 \times 1\text{ms})$

$$= 3 + 4999$$

$$= 5002\text{ms}$$

can be seen delay using segmentation
reduces significantly.

almost ($\frac{1}{5}$)

(d) Drawbacks:

① Packets must be reassembled in the correct order at the destination.

② Message segmentation generates many smaller packets. Since the header size remains constant regardless of packet size, segmentation increases the total no. of header bytes transmitted.

③ Packet loss: If even a single packet is lost during transmission, entire message may need to be retransmitted or reconstructed, leading to delays